

Data Flow Diagram

<i>Date</i>	29 September 2026
<i>Team ID</i>	UNM123APEXLAB
<i>Project Name</i>	ComicCraft - AI Comic Story Creator using Gemini
<i>Maximum Marks</i>	Models 2 Marks

Data Flow Diagram

This Data Flow Diagram shows how data moves through **ComicCraft**, a web-based application built with FastAPI that uses Google's Gemini AI models to generate personalized comic stories and illustrations from user-provided prompts. It shows the external entities (the User and the Gemini AI Models), the four main processes, and the data stores used to hold prompts, stories and generated images.

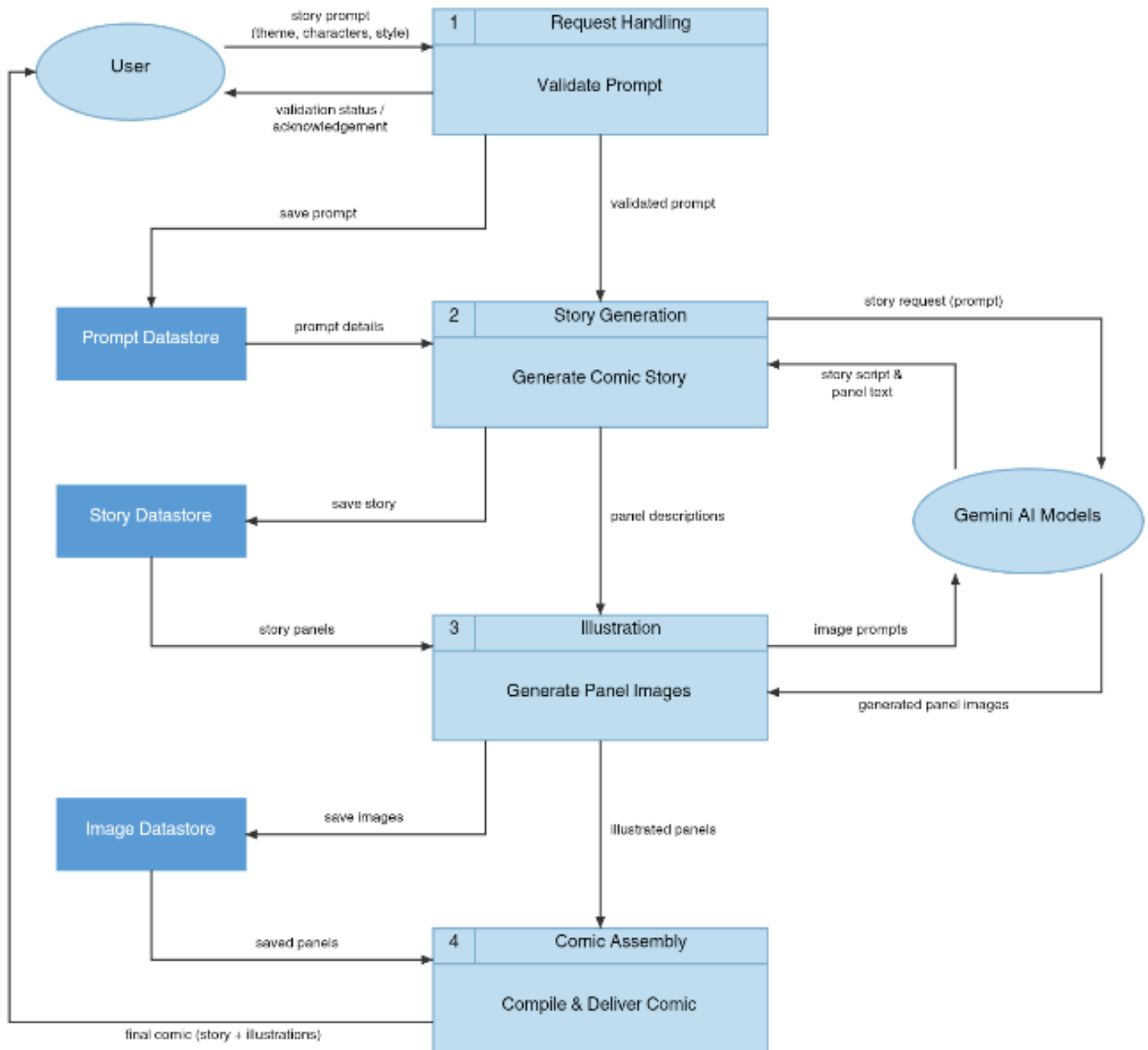
DFD Symbol Legend

<i>Symbol</i>	<i>Name</i>	<i>Description</i>
<i>Oval / Rounded shape</i>	<i>External Entity</i>	A person, organization, or system outside the project that sends or receives data (e.g., User, Gemini AI Models).
<i>Rectangle with numbered header</i>	<i>Process</i>	An activity that transforms incoming data into outgoing data (e.g., Validate Prompt, Generate Comic Story).
<i>Rectangle (solid fill, no number)</i>	<i>Data Store</i>	A place where data is held for later use (e.g., Prompt Datastore, Image Datastore).
<i>Labeled arrow</i>	<i>Data Flow</i>	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

ComicCraft - Data Flow Diagram

The complete data flow diagram of ComicCraft is shown on the next page, followed by a short description of each process, external entity and data store.

ComicCraft - Data Flow Diagram



Key:  External Entity  Process  Data Store  Data Flow

Figure: Data Flow Diagram of ComicCraft

Description of the Data Flow Diagram

Processes

<i>No.</i>	<i>Process</i>	<i>Description</i>
1	Request Handling - Validate Prompt	FastAPI receives the story prompt (theme, characters, art style) from the user, validates it, saves it in the Prompt Datastore and returns an acknowledgement.
2	Story Generation - Generate Comic Story	Sends the validated prompt to the Gemini text model, receives the story script and panel-wise text, and stores the story in the Story Datastore.
3	Illustration - Generate Panel Images	Builds image prompts from the story panels, sends them to the Gemini image model, receives the generated illustrations and saves them in the Image Datastore.
4	Comic Assembly - Compile & Deliver Comic	Combines the story text with the saved illustrations panel by panel and sends the final comic to the user.

External Entities and Data Stores

<i>Name</i>	<i>Type</i>	<i>Purpose</i>
User	External Entity	Provides the story prompt and receives the final comic.
Gemini AI Models	External Entity	Google's AI models that generate the story text and the comic illustrations.
Prompt Datastore	Data	Holds validated user prompts and their details.
Story Datastore	Store	Holds the generated story script and panel descriptions.
Image Datastore	Data	Holds the generated panel illustrations.

Store

Data

Store